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IMPACT OF FORIEGN INSTITUTIONAL INVESTMENT INFLOW AND OUTFLOW ON INDIAN STOCK MARKET RETURNS

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ABSTRACT:

This study try to evaluate the connection between FII inflows and outflows on Indian stock market returns. It analyse the link between FPI and work of major Indian equity indices like Sensex and Nifty. The research explores dynamics of FII movements from 2000 to 2026 period. It investigate patterns of capital flows market volatility and investor sentiment during this time. This work utilizes pattern analysis regression model and correlation analysis. Findings shows that although FII flows are positively corelated with equity place returns but relationship is very complicated and influenced by many things such as institutional investor behaviour domestic policy changes and global economic conditions.

The results reveal that FII outflows create much more strong market reactions than inflows. Cushioning effect of DIIs has become increasingly important in few years. With foreign ownership reaching 15 year low of 16.9% by September 2025 while domestic institutional participation has become stronger Indian equity market is seeing fundamental shift in its investor base.

Apart from helping policymakers regulators and industries to handle challenges of FII in coming markets this paper try to boost understanding about changing dynamics of capital flows in India.

Keywords: FII, Indian Stock Market, Capital Flows, Market Returns, Volatility, NSE, BSE, Sensex, Nifty.

1. Introduction

1.1 Background and Context

Since Indian economy opened up in 1991 foreign capital flows played very transformative role in shaping Indian financial markets. When equity markets opened for FIIs in September 1992 it became like major turning point. This allowed international investors to directly participate in India growth story. Over next thirty years FIIs emerged as very important players in Indian capital markets. They contributed alot in market liquidity price discovery and linking Indian markets with global financial world.

FIIs are basically different types of international entities who comes to emerging markets for investment opportunities. In India their presence grew very much. Their share in total foreign portfolio investment jumped from 47% in 1993-94 to nearly 74% by 2001-2002. But this growth path was never smooth. There were times of massive inflows and then sudden big capital outflows specially during 2008 global crisis and other periods of market volatility.

1.2 Evolution of FII Participation in India

FII investments in India faced many changes in legal rules since SEBI brought SEBI (FII) Regulations in 1995. These rules made complete guidelines for FII registration investment limits compliance and reporting. It created proper framework for foreign investment and also tried to protect market and investors. SEBI keeps changing and updating these regulations time to time according to market conditions and global practices. They also added KYC norms anti money laundering rules and measures to stop insider trading.

In recent years investment picture changed dramatically. By 2025 foreign portfolio investors took out almost 18 billion dollars from Indian shares. This was one of biggest outflow in recent times. By September 2025 foreign ownership in Indian equities fell to 15 year low of 16.9 % from 17.4 percent at end of 2024. This situation is completely opposite to Domestic Institutional Investors (DIIs) whose position became much stronger. Their mutual fund holdings alone reached 10.35 percent. This made total domestic institutional ownership go above FII levels for first time in more than ten years.

1.3 Research Problem and Significance

Big volatility and huge size of FII flows creates increased overview about their consequences on Indian stock place returns market stability investor confidence and complete money policy. It has become very central to

learn the link between FII movements and market performance for policymakers regulators and investors. This understanding helps in making better policies capital control and market development.

Studies on FII investment behaviour is very interesting. Many research shows markets react much more violently when FIIs withdraw money than when they bring money. This shows uneven reaction to foreign capital. Also FII money is very mobile. Even if India looks attractive foreign investors quickly run away during any short term risk or uncertainty. This behaviour increases market volatility.

1.4 Research Objectives

This work tries to achieve following objectives:

1. To check linking between FII coming in and outflows with Indian equity place returns on major indices like Nifty 50 and Sensex.
2. To examine how strong is correlation and statistical importance of FII flows on market volatility and returns.
3. To observe behavioural patterns how market reacts differently to FII coming in and outflows.
4. To analyse role of DIIs as stabilizing force during FII volatility periods.
5. To study changing FII-DII dynamics and impact on market stability and future policy implications.

2. Literature Review

2.1 Theoretical Framework of FPI

A. Theoretical base of FPI mostly comes from Portfolio Diversification Theory developed by Harry Markowitz. It says international investors try to divide their cash money across different countries markets having different correlation. Higher growth potential in emerging markets and their weak link with developed markets give good diversification chance. This thing was also torched by Berrill.

B. Andrew Lo's Adaptive Market Hypothesis (AMH) gives another framework to understand FII behaviour in emerging markets. According to Hiremath stock market efficiency keep changing with time. Investors findings for extra returns during market inefficiency periods. From this view FII flows react very fast to changing market conditions. They take benefit of inefficiencies when available and quickly pull out money when market look overpriced or better opportunities come somewhere else.

2.2 Determinants of Foreign Portfolio Investment in Emerging Markets

- A. FPI flows into emerging markets influenced by combination of global push factors and domestic pull factors. According to Calvo et al global variables like low interest rates and extra liquidity in developed economies greatly influence capital flows toward emerging nations. Similarly Fernandez-Arias (1996) emphasized that external conditions such as global risk and monetary policies of advanced economies strongly affect investment decisions.
- B. On domestic side Chuhan et al. (1998) found that economic growth market capitalization and good market returns are main things that determine FPI inflows. Portes and Rey also pointed importance of information availability financial market development and institutional quality to attract foreign investors. Stable macroeconomic conditions transparent rules and political stability further increase investor confidence and encourage continuous capital inflows.

3. Empirical Studies on FII Impact on Indian Markets

- A. Lot of empirical researches have examined correlation between FII movement and Indian equity place and results are mixed. Some correlation based work shows non negative linking between FII trading and major Indian stock indices. For example research by Gupta (2025) reported correlation of $r = 0.281$ for BSE Sensex and $r = 0.259$ for NSE Nifty suggesting somewhat positive link. These correlations aren't very strong. It means more other things affect market performance and FII flow is not only main driver. Still FII flows can act as leading indicator for guessing equity market changes and useful for investors and policymakers.
- B. Recent work from 2000 to 2024 also confirm importance of institutional flows. FII and DII flows have big impact on volatility of important indices like Nifty 50. These studies also point leverage impact which means non positive market returns increase volatility more sharply than positive returns of same size. This shows asymmetric nature of market response to FII movements.

4. Determinants of FII Flows to India

Understanding what actually drives FII investment decisions is very central for guessing their flows and making suitable policy responses. Regression analysis done for period 2011-2020 torched several statistically crucial reasons which determine FII flows into India.

- A. Domestic Market Performance: Sensex returns was strong predictor. This shows FIIs mostly follow momentum strategy and they increase investments when market is doing good.
- B. Exchange Rate Dynamics: Dollar to INR exchange rate is most influencing factor in FII flows.
- C. Interest Rate Differentials: FII flows gets affected by interest rates in USA and India. When US rates goes higher it often causes capital to flow back to America.
- D. Market Risk: FII flows gets negatively impacted by volatility in Indian market because risk avoider foreign investors always prefer stable environment for investment.

Further study conducted by Kaur and Dhillon searched that equity place risk domestic inflation and US interest rates puts negative pressure on FII investment. While Indian equity market returns market capitalization and open market policies have positive impact. This multifactor view highlights complexity of FII decision making which considers both local fundamentals and international financial situations.

5. Asymmetric Market Responses to FII Flows

Most interesting result in this work is that market reaction to FII inflows and outflows are asymmetric. Studies shows market reacts much more violently to FII withdrawals than to same size inflows. This thing points that when FIIs brings out money it creates panic in domestic investors and increases selling pressure which makes price fall even more sharply. On other hand FII inflows only gets moderate positive reaction maybe because local investors not so aggressive in buying when foreign money comes in.

7. The Evolving Role of DII

- A. Increasing importance of DIIs shows major structural shift in Indian equity market. Because they have better knowledge of local conditions DIIs shows more stability in long term investment than FIIs who have high capital mobility and very sensitive to global market conditions. Regulatory framework is also totally different FIIs have strict investment limits currently 24% of paid-up capital but DIIs have no such restrictions so they can put big value of domestic capital.
- B. Recent data points that FII to DII ownership ratio has gone below 1.0 and reached 0.98. This highlights growing domestic influence in Indian markets. This variation is very crucial for market stability as DII act like natural hedge against foreign capital volatility and helps absorb selling pressure during FII outflow periods.

8. Research Gaps

While existing studies give quite good information about FII and market return relationship but still several important gaps are there which need further investigation. First most of the studies only focus on old periods before 2020 and very little work has been done on post pandemic time and recent changes in FII-DII ownership pattern. Second we need more advanced type of analysis on non-linear relationships and threshold effects to see whether FII impact changes according to size and how long the flows continue. Third sector level analysis is still very poor. There is not much understanding how FIIs' preference for particular sectors specially heavy impact on IT stocks in recent times affects overall market behaviour.

9. Research Methodology

9.1 Research Design

This study uses quantitative research design. It combines correlation analysis regression modeling and trend analysis for investigating link between FII movement and equity place returns. The work follows positivist approach and try to find objective patterns and statistical relationships present in money market data.

9.2 Data Sources

The analysis is dependent on data from 2000 to March 2026 which covers more than twenty years of FII activity and place performance. This long period help to examine long term trends cyclical patterns and structural breaks in relationship between FII and market returns. Main data sources are:

- a) FII Flow Data: Daily and monthly FII investment numbers taken from SEBI NSDL and CDSL (2020)
- b) Market Return Data: Daily closing prices and returns of NSE Nifty 50 and BSE Sensex taken from NSE and BSE
- c) DII Flow Data: Domestic institutional investment figures from NSE and SEBI reports

Macroeconomic Variables: Exchange rates interest rates inflation and commodity prices collected from RBI and other global financial databases

9.3 Variables

A. Dependent Variables:

- a) Market Returns: Percentage returns of Nifty 50 and Sensex calculated using log returns method
- b) Market Volatility: Measured using standard deviation of daily returns in monthly windows and GARCH method for conditional volatility

B. Independent Variables:

- a) FII Net Investment: Net FII buying (total purchase minus total sales) measured in INR crores
- b) FII Investment as Percentage of Market Cap: FII flows compared to total market capitalization
- c) DII Net Investment: Net DII purchases used for comparison purpose

9.4 Analytical Techniques

A. Correlation Analysis: Pearson correlation values being calculated to see linear link between FII movement and place returns. Analysis check both current correlations and lagged correlations to find lead-lag effects.

B. Regression Analysis: Various regression structure are utilised to measure outcome of FII flows on market returns after controlling other factors.

9.5 Limitations

Several limitations are there in this study. First data availability and quality for early years (2000-2005) is not as good and complete as recent periods. Second this analysis mainly focus on total aggregate FII flows and have very less details about actual FII investment strategies like momentum or contrarian their small time or large term approach and sector preferences. Third endogeneity problem is always big challenge in money market research because market returns can affect FII movement and FII flows can also affect market returns at same time even when using Granger causality and instrumental variable techniques.

10. Results and Analysis

10.1 Descriptive Statistics and Trends

FII trends from 2000 to 2026 shows many big high and low due to different events. Like Lehman crisis of 2008-2009, quantitative easing period 2010-2013, Brexit in 2016 and COVID-19 in 2020 caused major swings

in FII activity in India. In 2025 FIIs sold almost 18 billion dollars of Indian stocks but Nifty 50 and Sensex did not crashed much because DII buying absorbed most of selling pressure.

10.2 Correlation Analysis Results

Results of regression and correlation analysis shows FII flows have moderate positive relationship with Indian equity place returns. Correlation for Nifty was 0.259 and for Sensex was 0.281. According to regression every 1000 crore net FII investment linked with 0.15 to 0.25 percent rise in index returns. But FII flows only explain market movements partly because DIIs company earnings exchange rate interest rates and other domestic factors also play very big roles.

10.3 The Changing FII-DII Dynamic

Most important finding is big shift in FII-DII ownership balance. Foreign ownership in Indian equities fell to 16.9 percent by September 2025 which is 15 year low. At same time DII holdings increased alot. FII to DII ownership ratio went below 1.0 and reached 0.98. This is historic thing because domestic institutional ownership now crossed foreign ownership for first time in modern Indian equity market history.

11. Discussion

Interpretation of Findings

Empirical results confirm that FII flows have big impact on equity place returns and changes but there are many nuances in this relationship. Positive but only moderate correlation (r around 0.25-0.28) between FII movement and demand indices shows FII is important but not dominant driver. This finding matches with efficient market theory which says asset prices reflect many informations and investor views and FII flow is only one part.

Unit directional Granger causality from FII movement to place returns supports that FIIs can better information or resources so they spot good opportunities before market fully prices them. This advantage may come from their advanced analysis global networks or quick capital movement.

Asymmetric market response to FII outflows versus inflows is very big behavioural finance thing. This pattern goes against efficient market hypothesis because it assumes symmetric price reaction. Instead it supports behavioural things like investor sentiment herding and loss aversion. Bigger panic during outflows shows losses hurt investors more than gains which matches with Kahneman and Tversky prospect theory.

12. Policy Implications

Study results have several important implications for policymakers and regulators:

12.1 Capital Flow Management

Although India mostly kept open capital account for portfolio investment but volatility of FII flows and their effect on place of sell stability may need some control measures during extreme stress times. But these measures should be made carefully so they don't damage investor confidence or India link with global markets. Temporary things like transaction tax or holding period during crisis can reduce bad short term flows while keeping medium term investment attractive.

12.2 Domestic Institutional Investor Development

Stabilizing DIIs during 2025 FII outflow shows importance of building strong domestic institutional capacity. Policies that help domestic institutions like tax benefits for long term saving regulatory changes for more diversification and better corporate governance for investor protection should get top priority.

12.3 Market Infrastructure Enhancement

SEBI framework for FIIs should keep changing to balance openness with stability. More disclosure requirements closer monitoring of big FII positions and better circuit breakers during high volatility can prevent big destabilizing moves while keeping India attractive for foreign capital.

12.4 Exchange Rate Policy Coordination

Because hard link between exchange rate and FII flows closer coordination between SEBI and RBI is must. Too much rupee volatility can start cycle of FII outflows and currency fall. So exchange rate management is important along with capital market rules.

13. Key Findings

The study establishes several key conclusions:

- a) Positive but moderate correlation: Correlation between FII flows and Indian stock market returns is weak between 0.25 to 0.28 but positive and statistically significant.
- b) FII flows lead market returns: Granger causality shows FII flows usually affect future market returns rather than just reacting to them.
- c) Asymmetric Responses: Due to panic selling herd behaviour and negative sentiment Indian market reacts much stronger to FII selling than to same size inflows.
- d) Volatility Impact: Market volatility greatly influenced by both FII and DII flows. Negative returns necessitates more changes than non negative returns of same size.
- e) Change in FII-DII Ownership: DIIs now play bigger stabilizing role by absorbing FII outflows and reducing dependence on foreign capital. FII ownership dropped to 16.9% by September 2025 and DII ownership crossed FII levels.

14. Policy Recommendations

Based on these understanding various policy suggestions come out:

- A. Strengthen Domestic Institution Capacity: Government should bring policies like tax breaks regulatory fluctuations and better investor protection to build strong domestic pension money assurance companies and mutual funds.
- B. Maintain Regulatory Balance: India should keep luring cross border investors but also take careful steps to control unstable short term flows specially during crisis periods.
- C. Enhance Market Infrastructure: To stop extreme volatility SEBI should make stronger position monitoring disclosure rules and circuit breaker systems.
- D. Coordinate Macroeconomic Policies: Better coordination needed between SEBI and RBI to manage connection between FII flows exchange rates and monetary policy.

15. Limitations and Future Research

This work has certain limitations which open doors for more research. First study only uses aggregate FII flows data. It can be improved by dividing investors according to their strategy investment horizon and type. Second

endogeneity problem in financial market research is always big challenge. It can be handled better with advanced methods like instrumental variables or natural experiment designs.

Future work can see into long term sustainability of DII led market support system. It should check whether domestic savings limits can stop future growth or DIIs can keep expanding further. Comparative studies can also be done to see if India DII growth model is unique or can be copied in other emerging markets.

16. Conclusion

Relationship between FII flows and Indian equity market returns is very complex and always changing. It gets influenced by both internal and external factors. Historically FII inflows had major impact on price discovery liquidity and short term market movements. Big inflows usually come during good market periods. But sudden outflows due to interest rate changes global risk or geopolitical issues can cause severe market corrections. This relationship is neither linear nor fixed. Macroeconomic factors monetary policy expectations currency stability and investor mood all affect how market reacts to FII activity.

In recent years increasing role of DIIs like mutual funds insurance companies and pension funds has caused structural shift in Indian equity market. This shift happened because of growing investor awareness government support for domestic asset management and financialization of household savings. Now Indian market become more strong against external shocks. Stable domestic inflows are now absorbing pressure which earlier caused long downturns due to FII withdrawals. This change marks fundamental shift from externally driven volatility to more balanced and internally supported market system.

Implications of this transition goes beyond short term market behaviour. Stronger domestic institutional base improves economic resilience by reducing dependence on volatile foreign capital. It also gives policymakers more autonomy because market stability becomes less sensitive to global liquidity cycles and external monetary tightening. Moreover diversified investor base reduces systemic risk and promotes long term investment horizon which strengthens overall financial system.

For policymakers main challenge is to find right balance. On one hand keeping openness to foreign capital is essential because FIIs bring capital global best practices governance standards and integration with international markets. On other hand strengthening domestic institutions is equally important for stability and reducing vulnerability to sudden capital flight. This needs continued rule support for financial togetherness

development of long term savings instruments and regulatory frameworks that encourage disciplined investment.

From investor point of view evolving dynamics give important lesson. While FII flows still remain relevant indicator of market sentiment they should not be seen in isolation or used as only basis for investment decisions. Short term market reactions to FII activity can often be misleading. Better investment approach should focus on company fundamentals earnings growth macroeconomic trends and growing role of domestic capital flows. Long term investors especially can benefit because market corrections due to temporary foreign outflows may actually create buying opportunities rather than risks.

Looking ahead as India continues on its economic development path interplay between foreign and domestic capital will keep changing. Factors like global interest rate cycles technological changes regulatory reforms and expansion of middle class will influence this balance. Continued deepening of domestic capital markets along with selective and stable foreign participation will likely define next phase of growth.

In conclusion changing relationship between FII flows and Indian stock market returns reflects big shift in national financial environment. Growing dominance of domestic institutions indicates increase in stability resilience and maturity of market. Understanding these dynamics is very central for policymakers who desire to make effective financial policies and for investors who desire to find opportunities in one of fastest growing major economies in world.

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